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Mutual inductance M_{21} is also defined as the opposing emf induced in the coil 2 when the rate of change of current through the coil 1 is $1As^{-1}$.

Similarly, if an electric current i_2 through coil 2 changes with time, then emf ε_1 is induced in coil 1. Therefore,

$$M_{12} = \frac{N_1\Phi_{12}}{i_2} \quad \text{and} \quad M_{12} = \frac{-\varepsilon_1}{di_2/dt}$$

where M_{12} is the mutual inductance of the coil 1 with respect to coil 2. It can be shown that for a given pair of coils, the mutual inductance is same.

$$\text{i.e.,} \quad M_{21} = M_{12} = M$$

In general, the mutual induction between two coils depends on size, shape, the number of turns of the coils, their relative orientation and permeability of the medium.

Unit of mutual-inductance

The unit of mutual inductance is also henry (H).

If $i_1 = 1A$ and $N_2\Phi_{21} = 1\ Wb\ turns$, then $M_{21} = 1H$.

Therefore, **the mutual inductance between two coils is said to be one henry if a current of $1A$ in coil 1 produces unit flux linkage in coil 2.**

If $di_1/dt = 1As^{-1}$ and $\varepsilon_2 = -1\ V$, then $M_{21} = 1H$.

Therefore, **the mutual inductance between two coils is one henry if a current changing at the rate of $1As^{-1}$ in coil 1 induces an opposing emf of $1V$ in coil 2.**

4.3.4 Mutual inductance between two long co-axial solenoids

Consider two long co-axial solenoids of same length l. The length of these solenoids is large when compared to their radii so that the magnetic field produced inside the solenoids is uniform and the fringing effect at the ends may be ignored. Let A_1 and A_2 be the area of cross section of the solenoids with A_1 being greater than A_2 as shown in Figure 4.23. The turn density of these solenoids are n_1 and n_2 respectively.

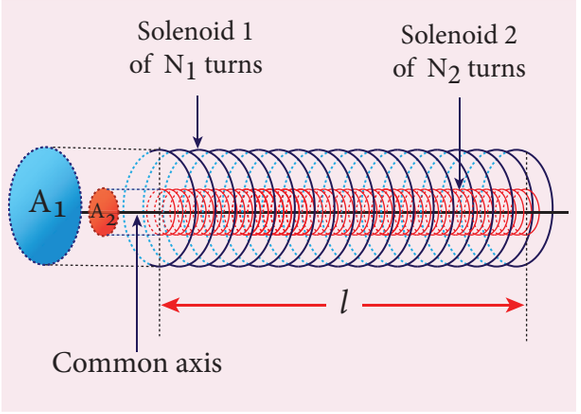


Figure 4.23 Mutual inductance of two long co-axial solenoids

Let i_1 be the current flowing through solenoid 1, then the magnetic field produced inside it is

$$B_1 = \mu_\circ\ n_1\ i_1$$

As the field lines of $\vec{B_1}$ are passing through the area bounded by solenoid 2, the magnetic flux is linked with each turn of solenoid 2 due to solenoid 1 and is given by

$$\Phi_{21} = \int_{A_2} \vec{B_1}.d\vec{A} = B_1A_2 \qquad \text{since } \theta = 0^\circ$$

$$= (\mu_\circ n_1 i_1)A_2$$

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